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Semester: - III

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Section: - E

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- ★ Question: - Implement NOT, AND, OR Gate using only NAND Gate in HDL. Draw the logical diagram for each and write Boolean expressions. Run the provide test scripts and capture the stimulation output.

- **HDL implementation for NOT Gate: -**

```
CHIP Not {  
    IN in;  
    OUT out;  
  
    PARTS:  
        Nand (a=in, b=in, out=out);  
}
```

- **Boolean expressions for NOT Gate: -**

For a NAND gate:

$$Y = A \cdot B$$

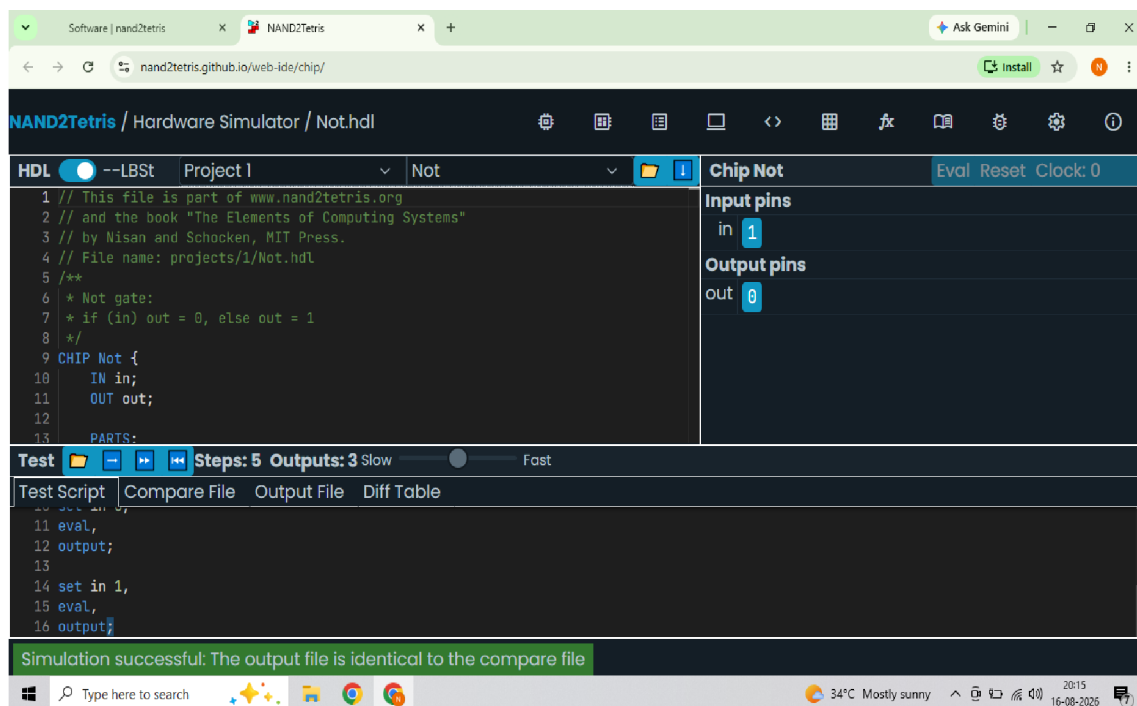
Since both inputs are connected to A:

$$Y = A \cdot A$$

Using the Boolean law $A \cdot A = A$:

$$Y = A$$

Therefore, the NAND gate behaves exactly like a **NOT gate**.



• HDL Implementation for AND Gate: -

```
CHIP And {
```

```
    IN a, b;
```

```
    OUT out;
```

```
    PARTS:
```

```
        Nand (a=a, b=b, out=nand1);
```

```
        Nand (a=nand1, b=nand1, out=out);
```

```
}
```

• Boolean Expression for AND Gate: -

First NAND gate:

$$X = \overline{A \cdot B}$$

Second NAND gate:

$$Y = \overline{X \cdot X}$$

Since $X \cdot X = X$:

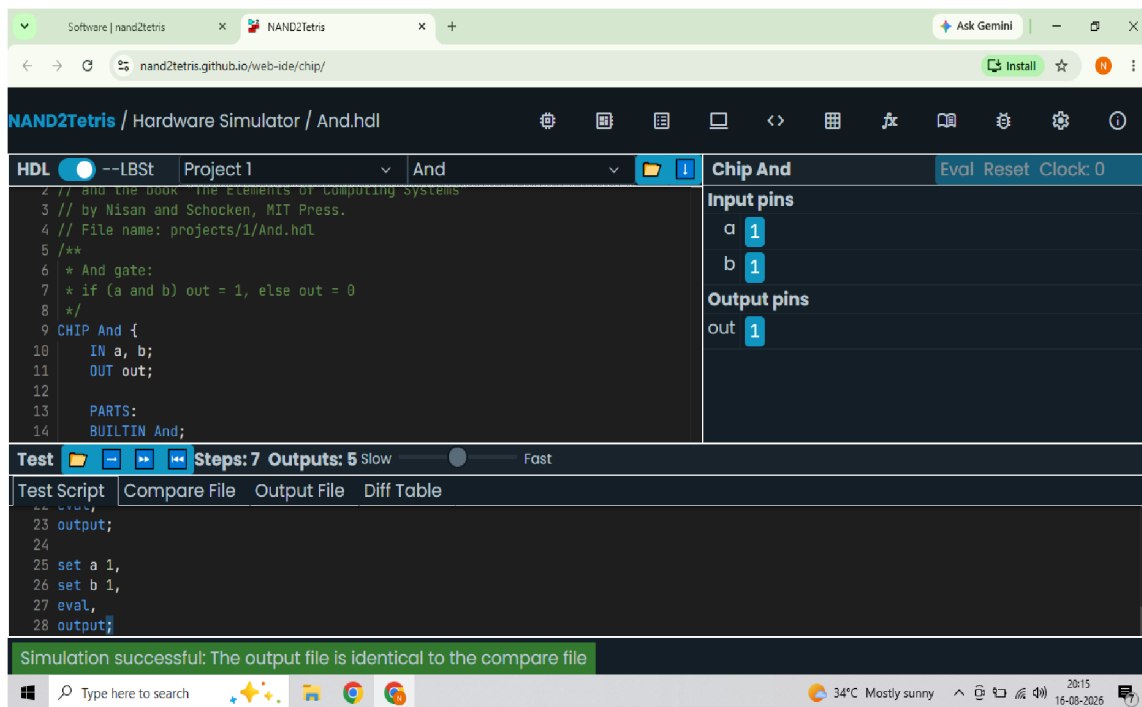
$$Y = \overline{X}$$

Substituting X:

$$Y = \overline{\overline{A \cdot B}}$$

Therefore:

$$Y = A \cdot B$$



- **HDL Implementation for OR Gate: -**

```
CHIP Or {
```

```
  IN a, b;
```

```
  OUT out;
```

```
  PARTS:
```

```
    Nand (a=a, b=a, out=not a);
```

```
    Nand (a=b, b=b, out=not b);
```

```
    Nand (a=not a, b=not b, out=out);
```

```
}
```

- **Boolean expressions for OR Gate: -**

$$OUT = \overline{X \cdot Y}$$

Substituting:

$$OUT = \overline{A} \cdot \overline{B}$$

Using **De Morgan's theorem**:

$$OUT = A + B$$

Therefore, the circuit works as an **OR gate**.

Software | nand2tetris

NAND2Tetris

Ask Gemini

nand2tetris.github.io/web-ide/chip/

NAND2Tetris / Hardware Simulator / Or.hdl

HDL

--LBSt

Project 1

Or

Chip Or

Eval

Reset

Clock: 0

```
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Or.hdl
5 /**
6  * Or gate:
7  * if (a or b) out = 1, else out = 0
8  */
9 CHIP Or {
10     IN a, b;
11     OUT out;
12
13     PARTS:
14         /// Replace this comment with your code.
15 }
```

Input pins

a 0

b 0

Output pins

out 0

Test

Steps: 0

Outputs: 0

Slow

Fast

Test Script

Compare File

Output File

Diff Table

```
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Or.tst
5
6 load Or.hdl,
7 compare-to Or.cmp,
8 output-list a b out;
```

HDL code: No syntax errors

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FIFA World Cup 2026...

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18-08-2026

